

## **Sid Mathur's Thematic ETF Screener – Relative-Strength Rotation**

2026-06-13

|                           |    |
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## Action Center

**As of 13-Jun-2026:** holding **23 ETFs** across **6 sectors** (of **2** currently in relative-LONG). Gross exposure **100%**, cash **0%**.

Full-period: **CAGR 11.5%**, **Sharpe 0.87**, **Sortino 1.20**, **Max DD -14.4%**, Calmar **0.79**. Signal on relative strength, P&L on the absolute ETFs held; no leverage; inverse-vol weighted within the selected leader set.

Rebalance is **monthly** (last trading day decision, next-morning execution). Signals are computed daily, but positions only change at month-end; intra-month the book is static. Min-hold 30 days is honored on each per-position basis.

## Latest rebal changes

Last rebalance: **01-Jun-2026** (24 trades).

## Entered

| ticker | name                             | sector                | weight |
|--------|----------------------------------|-----------------------|--------|
| VPL    | Vanguard FTSE Pacific ETF        | Japan Equity          | 3.8%   |
| XT     | iShares Future Exponential Te... | U.S. Large Cap Equity | 5.6%   |
| XNTK   | State Street SPDR NYSE Techno... | U.S. Large Cap Growth | 4.1%   |
| MTUM   | iShares MSCI USA Momentum Fac... | U.S. Large Cap Equity | 4.3%   |
| IWC    | iShares Microcap ETF             | U.S. Mid Cap Equity   | 4.2%   |
| IWO    | iShares Russell 2000 Growth F... | U.S. Mid Cap Equity   | 4.5%   |
| VTWG   | Vanguard Russell 2000 Growth ... | U.S. Mid Cap Equity   | 4.4%   |
| FLJH   | Franklin FTSE Japan Hedged ETF   | Japan Equity          | 5.5%   |
| DBJP   | Xtrackers MSCI Japan Hedged E... | Japan Equity          | 5.4%   |

## Exited

| ticker | name                             | sector              | prev_wt |
|--------|----------------------------------|---------------------|---------|
| RZV    | Invesco S&P SmallCap 600 Pure... | U.S. Mid Cap Equity | 8.3%    |

**Imminent moves****Next-in-line entries (eligible, in a leading sector, not yet held)**

| ticker | name                             | sector               | rs_momentum |
|--------|----------------------------------|----------------------|-------------|
| XTN    | State Street SPDR S&P Transpo... | U.S. Mid Cap Equity  | 15.1%       |
| JVAL   | JPMorgan U.S. Value Factor ETF   | U.S. Large Cap Value | 8.4%        |
| FTXR   | First Trust Nasdaq Transporta... | U.S. Mid Cap Equity  | 5.2%        |
| RZV    | Invesco S&P SmallCap 600 Pure... | U.S. Mid Cap Equity  | 5.2%        |
| RDVY   | First Trust Rising Dividend A... | U.S. Large Cap Value | 5.0%        |
| PAVE   | Global X U.S. Infrastructure ... | U.S. Large Cap Value | 4.5%        |
| RWJ    | Invesco S&P SmallCap 600 Reve... | U.S. Mid Cap Equity  | 3.9%        |
| IWS    | iShares Russell Mid-Cap Value... | U.S. Large Cap Value | 3.1%        |
| VONV   | Vanguard Russell 1000 Value E... | U.S. Large Cap Value | 3.0%        |
| SLYG   | State Street SPDR S&P 600 Sma... | U.S. Mid Cap Equity  | 2.9%        |

Performance vs benchmarks

Table 4: Same window. Defensive Strategy = dual-momentum clusters; Market = full-universe PC1 buy-and-hold.

| Metric   | Rel-Strength | Defensive Strategy | Market |
|----------|--------------|--------------------|--------|
| CAGR     | 11.5%        | 5.6%               | 7.4%   |
| Ann vol  | 12.6%        | 8.1%               | 15.1%  |
| Sharpe   | 0.87         | 0.67               | 0.55   |
| Sortino  | 1.20         | 0.94               | 0.78   |
| Max DD   | -14.4%       | -11.6%             | -26.1% |
| Calmar   | 0.79         | 0.48               | 0.28   |
| Hit rate | 56%          | 54%                | 53%    |

Table 5: Year-by-year.

| year | start_nav | end_nav | return | sharpe | max_drawdown | n_obs |
|------|-----------|---------|--------|--------|--------------|-------|
| 2021 | 100.0     | 100.0   | 0.0%   |        | 0.0%         | 74    |
| 2022 | 100.0     | 97.4    | -2.6%  | -0.58  | -4.1%        | 251   |
| 2023 | 97.3      | 95.7    | -1.7%  | -0.17  | -9.6%        | 250   |
| 2024 | 94.1      | 108.5   | 15.4%  | 0.94   | -10.9%       | 252   |
| 2025 | 108.0     | 141.5   | 31.0%  | 2.05   | -6.3%        | 250   |
| 2026 | 142.5     | 166.8   | 17.0%  | 1.52   | -14.4%       | 112   |

Monthly returns

Table 6: Monthly returns by year (live window).

|      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul  | Aug   | Sep   | Oct   | Nov   | Dec   |
|------|-------|-------|-------|-------|-------|-------|------|-------|-------|-------|-------|-------|
| 2021 |       |       |       |       |       |       |      |       | 0.0%  | 0.0%  | 0.0%  | 0.0%  |
| 2022 | 0.0%  | 0.0%  | 0.0%  | 0.0%  | 0.0%  | 0.0%  | 0.0% | 0.0%  | 0.0%  | 0.0%  | 0.0%  | -2.6% |
| 2023 | -0.1% | -2.9% | -0.5% | 0.4%  | -1.6% | 0.4%  | 3.2% | -4.2% | -1.1% | -1.0% | -0.1% | 6.0%  |
| 2024 | -3.1% | 5.9%  | 4.0%  | -4.5% | 3.7%  | -0.0% | 1.4% | 0.0%  | 2.4%  | 1.0%  | 8.6%  | -5.7% |
| 2025 | 2.3%  | 0.7%  | -0.3% | -0.1% | 0.0%  | 7.0%  | 1.3% | 5.3%  | 4.7%  | 2.4%  | 0.5%  | 3.4%  |
| 2026 | 7.4%  | 6.1%  | -9.0% | 6.4%  | 7.7%  | -0.7% |      |       |       |       |       |       |

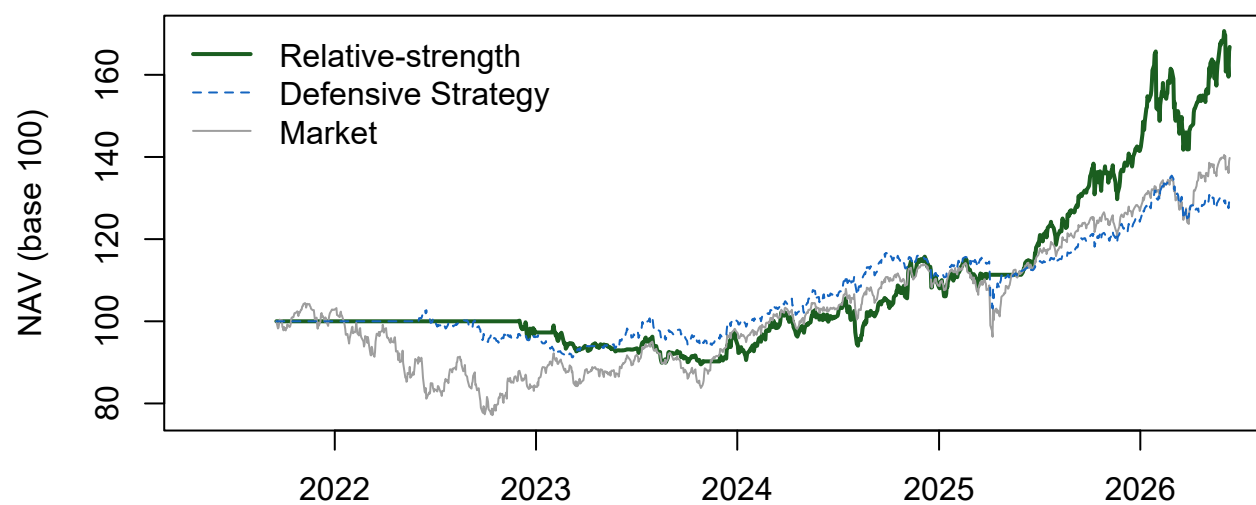


Figure 1: Growth of 100. Relative-strength rotation vs Defensive Strategy and market buy-and-hold.

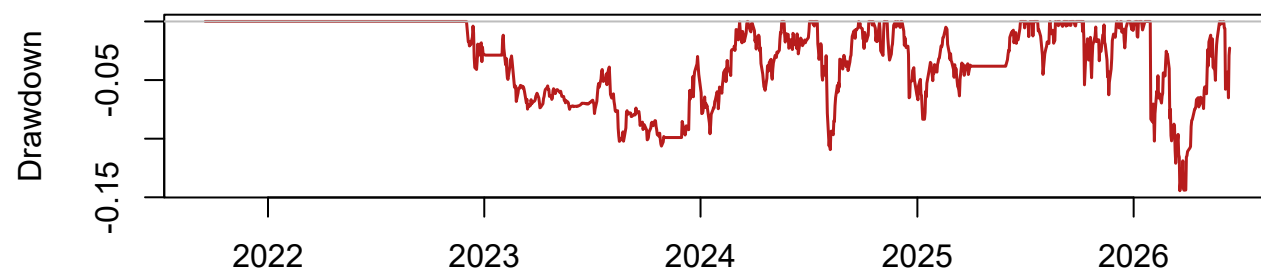


Figure 2: Drawdown of the relative-strength strategy (live window).

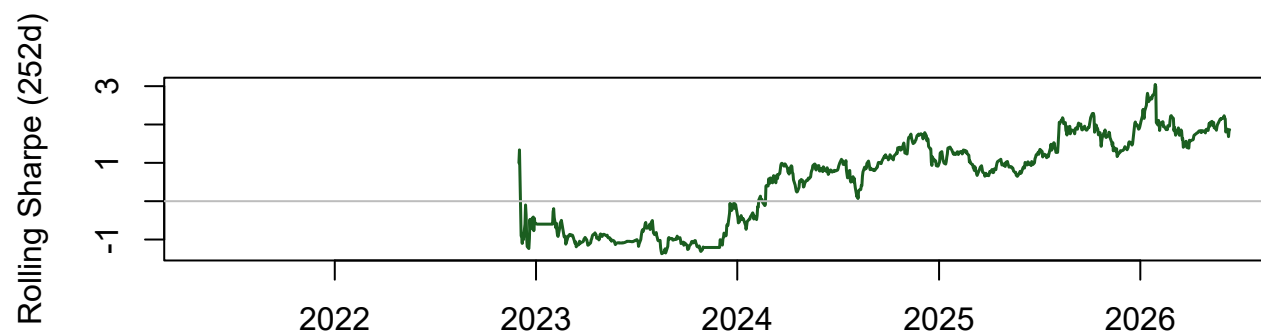


Figure 3: Rolling 252-day annualised Sharpe.



## Position Sheet

Current holdings, enriched with per-name vol, recent return, and the per-name signal state. Sorted by sector then weight desc.

Table 7: Current book. Vol-60d = trailing-60d annualised; ret-3m = 63-day cumulative; days held = calendar days in book.

| ticker | name                           | sector                  | weight | vol_60d | ret_3m | days_held | etf_rs | sector_rs |
|--------|--------------------------------|-------------------------|--------|---------|--------|-----------|--------|-----------|
| IDV    | iShares International Selec... | Developed ex-US Equity  | 6.7%   | 16.1%   | 5.8%   | 42        | FLAT   | FLAT      |
| FNDF   | Schwab Fundamental Internat... | Developed ex-US Equity  | 5.4%   | 21.9%   | 9.9%   | 42        | FLAT   | FLAT      |
| PXF    | Invesco RAFI Developed Mark... | Developed ex-US Equity  | 5.2%   | 22.7%   | 9.7%   | 42        | FLAT   | FLAT      |
| AAXJ   | iShares MSCI All Country As... | Emerging Markets Equity | 3.8%   | 33.8%   | 15.8%  | 42        | FLAT   | FLAT      |
| AIA    | iShares Asia 50 ETF            | Emerging Markets Equity | 3.1%   | 41.3%   | 23.8%  | 42        | LONG   | FLAT      |
| EWY    | iShares MSCI South Korea ETF   | Emerging Markets Equity | 1.7%   | 72.7%   | 39.2%  | 42        | FLAT   | FLAT      |
| FLJH   | Franklin FTSE Japan Hedged ... | Japan Equity            | 5.5%   | 20.6%   | 10.0%  | 11        | LONG   | FLAT      |
| DBJP   | Xtrackers MSCI Japan Hedged... | Japan Equity            | 5.4%   | 21.7%   | 10.0%  | 11        | LONG   | FLAT      |
| VPL    | Vanguard FTSE Pacific ETF      | Japan Equity            | 3.8%   | 31.7%   | 13.2%  | 11        | LONG   | FLAT      |
| XT     | iShares Future Exponential ... | U.S. Large Cap Equity   | 5.6%   | 22.2%   | 15.0%  | 11        | FLAT   | FLAT      |
| MTUM   | iShares MSCI USA Momentum F... | U.S. Large Cap Equity   | 4.3%   | 30.3%   | 30.1%  | 11        | LONG   | FLAT      |
| DWAS   | Invesco Dorsey Wright Small... | U.S. Large Cap Equity   | 4.2%   | 28.1%   | 20.7%  | 42        | LONG   | FLAT      |
| RPG    | Invesco S&P 500 Pure Growth... | U.S. Large Cap Equity   | 4.1%   | 29.8%   | 24.4%  | 42        | LONG   | FLAT      |
| ROBO   | ROBO Global Robotics and Au... | U.S. Large Cap Equity   | 3.6%   | 34.4%   | 15.1%  | 42        | FLAT   | FLAT      |
| XNTK   | State Street SPDR NYSE Tech... | U.S. Large Cap Growth   | 4.1%   | 34.0%   | 35.7%  | 11        | LONG   | FLAT      |
| IXN    | iShares Global Tech ETF        | U.S. Large Cap Growth   | 4.0%   | 33.9%   | 31.3%  | 42        | LONG   | FLAT      |
| QTUM   | Defiance Quantum ETF           | U.S. Large Cap Growth   | 3.5%   | 38.5%   | 42.5%  | 42        | LONG   | FLAT      |
| XSD    | State Street SPDR S&P Semic... | U.S. Large Cap Growth   | 2.4%   | 55.3%   | 74.7%  | 42        | LONG   | FLAT      |
| ISMD   | Inspire Small/Mid Cap ETF      | U.S. Mid Cap Equity     | 5.8%   | 19.0%   | 21.7%  | 42        | LONG   | LONG      |
| XSMO   | Invesco S&P SmallCap Moment... | U.S. Mid Cap Equity     | 4.7%   | 24.7%   | 17.8%  | 42        | FLAT   | LONG      |
| IWO    | iShares Russell 2000 Growth... | U.S. Mid Cap Equity     | 4.5%   | 26.0%   | 18.4%  | 11        | FLAT   | LONG      |
| VTWG   | Vanguard Russell 2000 Growt... | U.S. Mid Cap Equity     | 4.4%   | 26.2%   | 18.5%  | 11        | FLAT   | LONG      |
| IWC    | iShares Microcap ETF           | U.S. Mid Cap Equity     | 4.2%   | 28.4%   | 15.6%  | 11        | FLAT   | LONG      |

## Sector Overview

Table 8: All clusters, LONG first then FLAT; within each, RS-momentum desc. Held = positions taken; Members = ETFs in the cluster.

| sector                           | rs_state | rs_momentum | n_held | n_members |
|----------------------------------|----------|-------------|--------|-----------|
| U.S. Mid Cap Equity              | LONG     | 4.5%        | 5      | 71        |
| U.S. Large Cap Value             | LONG     | -0.3%       | 0      | 61        |
| U.S. Large Cap Growth            | FLAT     | 12.7%       | 4      | 56        |
| Emerging Markets Equity          | FLAT     | 5.4%        | 3      | 25        |
| Financials                       | FLAT     | 2.1%        | 0      | 14        |
| U.S. Large Cap Equity            | FLAT     | 1.8%        | 5      | 80        |
| Japan Equity                     | FLAT     | 1.2%        | 3      | 9         |
| Developed ex-US Equity           | FLAT     | -0.2%       | 3      | 61        |
| Real Estate                      | FLAT     | -1.4%       | 0      | 14        |
| Healthcare                       | FLAT     | -2.8%       | 0      | 17        |
| U.S. Dividend Equity             | FLAT     | -4.6%       | 0      | 25        |
| Materials                        | FLAT     | -5.9%       | 0      | 17        |
| Asia-Pacific Equity              | FLAT     | -6.5%       | 0      | 23        |
| High Yield Bond                  | FLAT     | -7.7%       | 0      | 21        |
| Bank Loan / Floating Rate        | FLAT     | -7.7%       | 0      | 9         |
| International Fixed Income       | FLAT     | -8.8%       | 0      | 6         |
| Municipal Bond                   | FLAT     | -9.0%       | 0      | 13        |
| Cash / Ultra-Short Fixed Income  | FLAT     | -9.0%       | 0      | 9         |
| Aggregate Fixed Income           | FLAT     | -9.1%       | 0      | 58        |
| Inflation-Protected Fixed Income | FLAT     | -9.8%       | 0      | 10        |
| Consumer Staples                 | FLAT     | -10.1%      | 0      | 11        |
| China Equity                     | FLAT     | -11.3%      | 0      | 15        |
| Broad Commodities                | FLAT     | -11.6%      | 0      | 10        |
| India Equity                     | FLAT     | -19.3%      | 0      | 11        |
| Gold                             | FLAT     | -24.3%      | 0      | 12        |
| Leveraged / Inverse              | FLAT     | -35.8%      | 0      | 2         |

Leading Cluster Details

Sector 2: U.S. Mid Cap Equity

- Relative-strength state: **LONG** RS momentum (63d): **4.5%** held: **5 / 71** members
- Theme: U.S. Mid Cap Equity



Currently held from this sector:

| ticker | name                           | weight | rs_mom | etf_rs | days_held |
|--------|--------------------------------|--------|--------|--------|-----------|
| ISMD   | Inspire Small/Mid Cap ETF      | 5.8%   | 7.0%   | LONG   | 42        |
| XSMO   | Invesco S&P SmallCap Moment... | 4.7%   | 3.8%   | FLAT   | 42        |
| IWO    | iShares Russell 2000 Growth... | 4.5%   | 4.6%   | FLAT   | 11        |
| VTWG   | Vanguard Russell 2000 Growt... | 4.4%   | 4.6%   | FLAT   | 11        |
| IWC    | iShares Microcap ETF           | 4.2%   | 2.4%   | FLAT   | 11        |

Strategy-1 replication basket (context):

| ticker | name                         | rep_wt | held_now |
|--------|------------------------------|--------|----------|
| IJH    | iShares Core S&P Mid-Cap ETF | 101.8% |          |

Sector 3: U.S. Large Cap Value

- Relative-strength state: **LONG** RS momentum (63d): **-0.3%** held: **0 / 61** members
- Theme: U.S. Large Cap Value



No member of this sector currently passes both the relative-LONG and the absolute overlay.

Strategy-1 replication basket (context):

| ticker | name               | rep_wt | held_now |
|--------|--------------------|--------|----------|
| VTV    | Vanguard Value ETF | 109.8% |          |

Sector 1: U.S. Large Cap Equity

- Relative-strength state: **FLAT** RS momentum (63d): **1.8%** held: **5 / 80** members
- Theme: U.S. Large Cap Equity



Currently held from this sector:

| ticker | name                           | weight | rs_mom | etf_rs | days_held |
|--------|--------------------------------|--------|--------|--------|-----------|
| XT     | iShares Future Exponential ... | 5.6%   | 4.3%   | FLAT   | 11        |
| MTUM   | iShares MSCI USA Momentum F... | 4.3%   | 18.3%  | LONG   | 11        |
| DWAS   | Invesco Dorsey Wright Small... | 4.2%   | 10.1%  | LONG   | 42        |
| RPG    | Invesco S&P 500 Pure Growth... | 4.1%   | 13.6%  | LONG   | 42        |
| ROBO   | ROBO Global Robotics and Au... | 3.6%   | 5.3%   | FLAT   | 42        |

Strategy-1 replication basket (context):

| ticker | name                 | rep_wt | held_now |
|--------|----------------------|--------|----------|
| V00    | Vanguard S&P 500 ETF | 101.0% |          |

Sector 7: Emerging Markets Equity

- Relative-strength state: **FLAT** RS momentum (63d): **5.4%** held: **3 / 25** members
- Theme: Emerging Markets Equity



Currently held from this sector:

| ticker | name                           | weight | rs_mom | etf_rs | days_held |
|--------|--------------------------------|--------|--------|--------|-----------|
| AAXJ   | iShares MSCI All Country As... | 3.8%   | 3.5%   | FLAT   | 42        |
| AIA    | iShares Asia 50 ETF            | 3.1%   | 12.2%  | LONG   | 42        |
| EWY    | iShares MSCI South Korea ETF   | 1.7%   | 32.5%  | FLAT   | 42        |

Strategy-1 replication basket (context):

| ticker | name                           | rep_wt | held_now |
|--------|--------------------------------|--------|----------|
| IEMG   | iShares Core MSCI Emerging ... | 92.5%  |          |

Sector 4: Developed ex-US Equity

- Relative-strength state: **FLAT** RS momentum (63d): **-0.2%** held: **3 / 61** members
- Theme: Developed ex-US Equity



Currently held from this sector:

| ticker | name                           | weight | rs_mom | etf_rs | days_held |
|--------|--------------------------------|--------|--------|--------|-----------|
| IDV    | iShares International Selec... | 6.7%   | -2.1%  | FLAT   | 42        |
| FNDF   | Schwab Fundamental Internat... | 5.4%   | 2.3%   | FLAT   | 42        |
| PXF    | Invesco RAFI Developed Mark... | 5.2%   | 2.3%   | FLAT   | 42        |

Strategy-1 replication basket (context):

| ticker | name                           | rep_wt | held_now |
|--------|--------------------------------|--------|----------|
| VEA    | Vanguard FTSE Developed Mar... | 96.7%  |          |

Sector 22: Japan Equity

- Relative-strength state: **FLAT** RS momentum (63d): **1.2%** held: **3 / 9** members
- Theme: Japan Equity



Currently held from this sector:

| ticker | name                           | weight | rs_mom | etf_rs | days_held |
|--------|--------------------------------|--------|--------|--------|-----------|
| FLJH   | Franklin FTSE Japan Hedged ... | 5.5%   | 0.4%   | LONG   | 11        |
| DBJP   | Xtrackers MSCI Japan Hedged... | 5.4%   | 0.8%   | LONG   | 11        |
| VPL    | Vanguard FTSE Pacific ETF      | 3.8%   | 5.5%   | LONG   | 11        |

Strategy-1 replication basket (context):

| ticker | name                           | rep_wt | held_now |
|--------|--------------------------------|--------|----------|
| EWJ    | iShares MSCI Japan Index Fu... | 90.9%  |          |



Sector 6: U.S. Large Cap Growth

- Relative-strength state: **FLAT** RS momentum (63d): **12.7%** held: **4 / 56** members
- Theme: U.S. Large Cap Growth



Currently held from this sector:

| ticker | name                           | weight | rs_mom | etf_rs | days_held |
|--------|--------------------------------|--------|--------|--------|-----------|
| XNTK   | State Street SPDR NYSE Tech... | 4.1%   | 11.9%  | LONG   | 11        |
| IXN    | iShares Global Tech ETF        | 4.0%   | 8.9%   | LONG   | 42        |
| QTUM   | Defiance Quantum ETF           | 3.5%   | 18.2%  | LONG   | 42        |
| XSD    | State Street SPDR S&P Semic... | 2.4%   | 46.0%  | LONG   | 42        |

Strategy-1 replication basket (context):

| ticker | name                        | rep_wt | held_now |
|--------|-----------------------------|--------|----------|
| QQQ    | Invesco QQQ Trust, Series 1 | 103.6% |          |

## Closed Trades

The strategy rebalances **monthly** – signals are computed daily, but positions only change at month-end (last trading day decision, execution next morning). The closed-trade ledger below is derived from the *engine's actual book*: each row is a position that the book opened and later closed, with hold duration measured from when the name first appeared in the weights to when it dropped out.

Closed positions: **371**. Mean hold: **44 days**. Median hold: **30 days**. Mean gross return: **2.0%** (absolute price over the held window).

Win rate: **58%**.

Table 22: Top 10 winning closed positions (engine-derived).

| ticker | entry      | exit       | hold_days | ret   |
|--------|------------|------------|-----------|-------|
| COPX   | 2025-12-01 | 2026-02-27 | 88        | 47.2% |
| CNXT   | 2025-08-01 | 2025-10-31 | 91        | 37.7% |
| GDXJ   | 2025-07-01 | 2025-10-31 | 122       | 34.6% |
| GDX    | 2025-08-01 | 2025-10-31 | 91        | 34.5% |
| ARKQ   | 2025-08-01 | 2025-10-31 | 91        | 29.9% |
| ARKW   | 2025-06-02 | 2025-08-29 | 88        | 27.2% |
| COPX   | 2025-09-02 | 2025-10-31 | 59        | 21.9% |
| SIL    | 2025-07-01 | 2025-08-29 | 59        | 19.4% |
| BLOK   | 2025-06-02 | 2025-07-31 | 59        | 19.3% |
| SMH    | 2024-02-01 | 2024-03-28 | 56        | 19.0% |

Table 23: Top 10 losing closed positions (engine-derived).

| ticker | entry      | exit       | hold_days | ret    |
|--------|------------|------------|-----------|--------|
| BLOK   | 2024-03-01 | 2024-04-30 | 60        | -11.1% |
| URA    | 2026-02-02 | 2026-03-31 | 57        | -11.4% |
| GDX    | 2023-12-01 | 2024-01-31 | 61        | -11.4% |
| RFV    | 2023-02-01 | 2023-03-31 | 58        | -12.4% |
| RZV    | 2023-02-01 | 2023-03-31 | 58        | -12.9% |
| GDXJ   | 2023-12-01 | 2024-01-31 | 61        | -14.5% |
| EZA    | 2026-03-02 | 2026-03-31 | 29        | -16.2% |
| PSI    | 2023-09-01 | 2023-10-31 | 60        | -19.2% |
| GDX    | 2026-03-02 | 2026-03-31 | 29        | -21.9% |
| GOAU   | 2026-03-02 | 2026-03-31 | 29        | -22.3% |

## ETF RS Leaders &amp; Laggards

The relative-strength analogue of strategy 1's Net Signal Strength: ETFs ranked by their **63-day relative-strength momentum** (ETF vs its own cluster PC1). Bullish list is restricted to ETFs in clusters currently in relative-LONG; bearish list is unrestricted.

Table 24: Top 20 most bullish ETFs by RS momentum (restricted to leading sectors).

| ticker | name                             | sector               | rs_mom | held | etf_rs |
|--------|----------------------------------|----------------------|--------|------|--------|
| VLUE   | iShares MSCI USA Value Factor... | U.S. Large Cap Value | 24.7%  |      | FLAT   |
| XTN    | State Street SPDR S&P Transpo... | U.S. Mid Cap Equity  | 15.1%  |      | LONG   |
| JVAL   | JPMorgan U.S. Value Factor ETF   | U.S. Large Cap Value | 8.4%   |      | LONG   |
| ETHO   | Amplify Etho Climate Leadersh... | U.S. Large Cap Value | 8.2%   |      | FLAT   |
| ISMD   | Inspire Small/Mid Cap ETF        | U.S. Mid Cap Equity  | 7.0%   | Y    | LONG   |
| JETS   | U.S. Global Jets ETF             | U.S. Mid Cap Equity  | 5.5%   |      | FLAT   |
| SCHA   | Schwab U.S. Small-Cap ETF        | U.S. Mid Cap Equity  | 5.3%   |      | FLAT   |
| FTXR   | First Trust Nasdaq Transporta... | U.S. Mid Cap Equity  | 5.2%   |      | LONG   |
| RZV    | Invesco S&P SmallCap 600 Pure... | U.S. Mid Cap Equity  | 5.2%   |      | LONG   |
| RDVY   | First Trust Rising Dividend A... | U.S. Large Cap Value | 5.0%   |      | LONG   |
| VTWG   | Vanguard Russell 2000 Growth ... | U.S. Mid Cap Equity  | 4.6%   | Y    | FLAT   |
| IWO    | iShares Russell 2000 Growth F... | U.S. Mid Cap Equity  | 4.6%   | Y    | FLAT   |
| PAVE   | Global X U.S. Infrastructure ... | U.S. Large Cap Value | 4.5%   |      | LONG   |
| IMCB   | iShares Morningstar Mid-Cap E... | U.S. Large Cap Value | 3.9%   |      | FLAT   |
| RWJ    | Invesco S&P SmallCap 600 Reve... | U.S. Mid Cap Equity  | 3.9%   |      | LONG   |
| XSMO   | Invesco S&P SmallCap Momentum... | U.S. Mid Cap Equity  | 3.8%   | Y    | FLAT   |
| FYX    | First Trust Small Cap Core Al... | U.S. Mid Cap Equity  | 3.7%   |      | FLAT   |
| PRF    | Invesco RAFI US 1000 ETF         | U.S. Large Cap Value | 3.7%   |      | FLAT   |
| VTWO   | Vanguard Russell 2000 ETF        | U.S. Mid Cap Equity  | 3.3%   |      | FLAT   |
| FEX    | First Trust Large Cap Core Al... | U.S. Large Cap Value | 3.2%   |      | FLAT   |

Table 25: Top 20 most bearish ETFs by RS momentum (full universe).

| ticker | name                             | sector                | rs_mom  | sector_rs |
|--------|----------------------------------|-----------------------|---------|-----------|
| PPLT   | abrdn Physical Platinum Share... | India Equity          | -210.1% | FLAT      |
| NOBL   | ProShares S&P 500 Dividend Ar... | India Equity          | -61.4%  | FLAT      |
| SQQQ   | ProShares UltraPro Short QQQ     | Leveraged / Inverse   | -28.0%  | FLAT      |
| PNQI   | Invesco Nasdaq Internet ETF      | U.S. Large Cap Growth | -21.0%  | FLAT      |
| FINX   | Global X FinTech ETF             | U.S. Large Cap Growth | -20.1%  | FLAT      |
| IPAY   | Amplify Digital Payments ETF     | U.S. Large Cap Growth | -19.5%  | FLAT      |
| IHI    | iShares U.S. Medical Devices ... | Healthcare            | -15.6%  | FLAT      |
| QYLD   | Global X NASDAQ 100 Covered C... | U.S. Large Cap Growth | -15.4%  | FLAT      |
| IBUY   | Amplify Online Retail ETF        | U.S. Large Cap Growth | -15.4%  | FLAT      |
| ESPO   | VanEck Video Gaming and eSpor... | U.S. Large Cap Equity | -15.2%  | FLAT      |

|      |                                  |                         |        |      |
|------|----------------------------------|-------------------------|--------|------|
| IGV  | iShares Expanded Tech-Softwar... | U.S. Large Cap Growth   | -14.7% | FLAT |
| PTLC | Pacer Trendpilot US Large Cap... | U.S. Large Cap Growth   | -14.4% | FLAT |
| ARKK | ARK Innovation ETF               | U.S. Large Cap Growth   | -14.1% | FLAT |
| XLC  | State Street Communication Se... | U.S. Large Cap Equity   | -13.0% | FLAT |
| URA  | Global X Uranium ETF             | Materials               | -13.0% | FLAT |
| SDIV | Global X SuperDividend ETF       | Emerging Markets Equity | -12.2% | FLAT |
| FDN  | First Trust DJ Internet Index... | U.S. Large Cap Growth   | -11.9% | FLAT |
| ARKQ | ARK Autonomous Technology & R... | U.S. Large Cap Growth   | -11.9% | FLAT |
| EIDO | iShares MSCI Indonesia ETF       | India Equity            | -11.8% | FLAT |
| SMMV | iShares MSCI USA Small-Cap Mi... | U.S. Mid Cap Equity     | -11.6% | LONG |

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## Watchlist

Table 26: Eligible leaders in leading sectors not currently held (next in line at next rebal).

| ticker | name                             | sector               | rs_mom |
|--------|----------------------------------|----------------------|--------|
| XTN    | State Street SPDR S&P Transpo... | U.S. Mid Cap Equity  | 15.1%  |
| JVAL   | JPMorgan U.S. Value Factor ETF   | U.S. Large Cap Value | 8.4%   |
| FTXR   | First Trust Nasdaq Transporta... | U.S. Mid Cap Equity  | 5.2%   |
| RZV    | Invesco S&P SmallCap 600 Pure... | U.S. Mid Cap Equity  | 5.2%   |
| RDVY   | First Trust Rising Dividend A... | U.S. Large Cap Value | 5.0%   |
| PAVE   | Global X U.S. Infrastructure ... | U.S. Large Cap Value | 4.5%   |
| RWJ    | Invesco S&P SmallCap 600 Reve... | U.S. Mid Cap Equity  | 3.9%   |
| IWS    | iShares Russell Mid-Cap Value... | U.S. Large Cap Value | 3.1%   |
| VONV   | Vanguard Russell 1000 Value E... | U.S. Large Cap Value | 3.0%   |
| SLYG   | State Street SPDR S&P 600 Sma... | U.S. Mid Cap Equity  | 2.9%   |
| VIOG   | Vanguard S&P Small-Cap 600 Gr... | U.S. Mid Cap Equity  | 2.9%   |
| PSC    | Principal U.S. Small-Cap ETF     | U.S. Mid Cap Equity  | 2.7%   |
| SCHV   | Schwab U.S. Large-Cap Value E... | U.S. Large Cap Value | 2.6%   |
| IYT    | iShares U.S. Transportation E... | U.S. Mid Cap Equity  | 2.5%   |
| SPSM   | State Street SPDR Portfolio S... | U.S. Mid Cap Equity  | 2.3%   |

**Run Parameters**

| parameter                       | value        |
|---------------------------------|--------------|
| Holding mode                    | etf_topk     |
| Signal (short/buy/sell MA)      | 5 / 50 / 20  |
| Sell cushion k                  | 0.5          |
| Min hold (days)                 | 30           |
| Rearm blocked buys              | TRUE         |
| RS ranking window               | 63           |
| Leading clusters (n)            | 6            |
| Leaders per cluster (top_k)     | 3            |
| Weight scheme                   | inverse_vol  |
| Vol window (for inverse-vol)    | 60           |
| Per-name cap                    | 20%          |
| Relative entry filter (abs-mom) | 252d         |
| Absolute overlay (abs-mom)      | 252d         |
| Market gate                     | ON (200d MA) |
| Rebalance                       | monthly      |
| Cost (bps)                      | 10           |